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Search prominence and return costs☆

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ABSTRACT

It is widely held that sellers prefer to appear early in a consumer's search, but evidence suggests this need not be the case. We develop a model which incorporates costly search and costly return and demonstrate that appearing later may be better. When return is free, prominence is desirable by standard logic, however costly return induces a tradeoff – it benefits an earlier seller by reducing the initial search but also benefits a later seller by preventing return conditional on search. We show that for small search costs later is better whenever high outcomes have a low likelihood, or whenever two independent match value draws are likely to be near one another. Later can still be better if sellers compete in prices prior to search. Finally, with many sellers the optimal position may be first, last, or in between but earlier positions are favored as the number of sellers grows.

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1. Introduction

A job candidate schedules an interview with an employer and is aware of the dates other candidates will be interviewed. Which date should she request? A car dealership can

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open a location at the center of town where consumers are likely to start their search, or on the outskirts where consumers come only after visiting more centrally located competitors. Which location is optimal? A newspaper provides content to an aggregator, where each reader inspects a list of competing article titles and accompanying blurbs in sequence and chooses an article to read in more detail. What is the newspaper's optimal position on this list?

Conventional wisdom is that in all such situations in which buyers choose from different options on a list, position matters and earlier positions are better than those further down. And indeed there is considerable empirical support for this intuition. For example, there is abundant evidence that earlier position is an important determinant of the clicks sellers receive from price comparison sites, auction sites, and online market places (See Ansari and Mela, 2003; Ellison and Ellison, 2009; Baye et al., 2009; Brynjolfsson et al., 2010, and the studies cited therein).

Explanations for such top position effects often rely on behavioral arguments (e.g. Dayan and Bar-Hillel, 2011; Murphy et al., 2006). An alternative is the explanation provided by Armstrong et al. (2009) and Zhou (2011), where consumers engage in costly sequential search. Consumers follow a threshold rule, and consequently those sellers searched earlier gain a competitive advantage whenever their offer exceeds this threshold, allowing them to sell to consumers who would otherwise have purchased from later competitors. The theory thus predicts not only that first is best, but that across all positions earlier is better.

However, there is also evidence that sometimes it is more desirable to appear further down on the list. For example, Novarese and Wilson (2013) analyze position effects for the number of hits and downloads of REPEC economics working papers on lists emailed to academic subscribers. While the authors confirm the existence of a strong top position bias, they also find a bias favoring working papers at the bottom of the list, relative to items in the position immediately above them. Feenberg et al. (2017) find an even stronger bottom position effect for downloads of NBER working papers and shows that "when conditioning on rank, the effect of being listed last is not significantly different from the effect of being listed first".

In the context of online hotel listings, De los Santos and Koulayev (2013) find a still greater variety of position effects, showing that sometimes even middle positions can be better than higher up or higher down positions. Documenting the number of clicks for search results for hotels on Kayak.com listed in decreasing popularity, they find that while the click rate decreases within each page of results, the monotonicity is broken across pages. For instance, the 16th result which appears at the top of page two received more clicks than any of the results 4–15 on the first page, and similarly the 31st result which appears at the top of the third page received more clicks than any of the results 20–30 on the second page.

To address this richer body of evidence about position effects, we develop a more comprehensive model of search position to accommodate both settings in which earlier is better as well as settings in which later is better. In particular, we investigate the effect of

search position by incorporating costly return into the standard differentiated products search framework of Wolinsky (1986). While return costs are often present in reality, they are typically assumed away in the search literature for reasons of tractability. Here we show that for the question of search position they play as important a role as search costs, and thus must be included.

We begin by considering a duopoly, where our analysis pivots around a basic tradeoff between two opposing effects of return costs. In particular, being searched earlier is advantageous for a seller when its prospective customers draw a relatively high utility value, because then search and return costs deter further search. On the other hand, because return is costly, consumers who draw a low or intermediate value at an earlier seller and keep searching are then deterred from going back even when a later seller proves to be no better or even slightly worse. We characterize conditions when the net outcome of this tradeoff benefits the earlier seller and when it benefits the later seller. Quite intuitively, later is better when high outcomes are unlikely, so that consumers search past the prominent seller, and when draws tend to be similar across sellers, thus making the return cost pivotal.

To study the monotonicity properties of search order we then extend the analysis to an N-seller oligopoly. Here a new dynamic emerges, owing to the fact that with return costs the search strategy is not stationary, as demonstrated by Janssen and Parakhonyak (2014), and instead consumers become less selective as they work down the list. Consequently, given the decreasing sequence of utility thresholds $(\hat{u}_1, \hat{u}_2, \ldots, \hat{u}_{N-1})$, seller 1 has an advantage over realizations above $\hat{u}_1$, every interior seller i has an advantage over realizations in the interval $(\hat{u}_i, \hat{u}_{i-1})$, and the final seller N has an advantage over all realizations below $\hat{u}_{N-1}$. We show that depending on the utility distribution it may be optimal to be first, last, or in between, but that as more sellers are added first improves not only relative to last but also relative to all the intermediate positions, matching the empirical findings of Novarese and Wilson (2013) and others.

Our paper is related to Armstrong and Zhou (2015), who consider a seller that can impose an infinite return cost by making an exploding offer. In their setting earlier sellers always have an advantage: either costly return aids the earlier seller in which case she makes an exploding offer, or costly return harms the earlier seller in which case she allows consumers to return at no cost and still benefits from the advantage of the search cost. By contrast, we determine when an earlier seller is at a disadvantage and hence later is better in the presence of exogenous return costs.

The paper proceeds as follows. We begin in Section 2 with a duopoly example with a simple binomial distribution and demonstrate that later can be better whenever the high outcome is sufficiently unlikely. In Section 3 we then extend the duopoly analysis to general distributions of match values, describe the tradeoff associated with search order, and derive distributional conditions under which earlier or later is better. In Section 4 we add to the model a first stage pricing game, and show that later can still be better, although the presence of the pricing stage harms the later seller due to issues related to commitment. Then in Section 5 we analyze the setting with $N \geq 3$ sellers and demonstrate

that first, last, or in between can be best due to the fact that consumers become less selective during search. Section 6 then concludes.

2. An example

A consumer with unit demand may buy from seller A or seller B. The consumer's utility for either seller i's product is uncertain and given by

$$u_i = \begin{cases} v & \text{with probability } 1-x \\ v+\delta & \text{with probability } x \end{cases},$$

with u_A and u_B drawn independently. The consumer does not initially know her utilities but may learn them by searching sequentially, first the prominent seller A and then the follower seller B, with search cost $s \in [0,\, x\delta]$ for each visit. In addition there is a recall cost $r > 0$ if the consumer wishes to return and accept at A after having visited B. The consumer has no outside option and must buy from either A or B.

It is optimal for the consumer to buy immediately if $u_A = v + \delta$ and to search and never return if $u_A = v$. The decision to not return is optimal because the second offer is at least as good as the first, and the search decision is optimal because the benefit of search outweighs the cost by assumption ($x\delta \geq s$). Consequently, the consumer buys from A with probability x and from B with probability $1 - x$, and B makes the sale more often whenever $x < \frac{1}{2}$.

The tradeoff to prominence is most easily seen relative to the informed benchmark, in which the consumer buys the better of the two products and buys either with equal chance in case of ties. In the current example with search frictions, the consumer still buys the better of the two products when one's value is high and the other's is low. However, the search cost breaks high-outcome ties in favor of the prominent firm and low-outcome ties in favor of the follower. Imposing that return costs are zero, as is typically done in the literature, negates the second effect and makes prominence unambiguously attractive. However, in the present example if $x < \frac{1}{2}$ then it is better to be second for *any* positive return cost.

In its simplicity this example clearly demonstrates the tradeoff associated with prominence but also negates some other important dynamics. For instance, due to the fact that there are only two outcomes the consumer never returns regardless of the cost, as long as it is strictly positive. An increase in r thus has no effect on either the search or return decision, but in general one may expect that a higher return cost both reduces return conditional on search and, for this very reason, reduces the likelihood of search in the first place. It is then not immediately clear whether the prominent firm is helped or harmed by an increased return cost. In addition, that later is better in bimodal distributions with a lower probability x of the high outcome relies partly on the fact that as the distribution changes the search decision remains unchanged. Again this is not the case in general, and one may wonder what are the distributional properties more generally that

favor the prominent firm or the follower? To answer these and other questions, we now consider a richer setting and derive the conditions under which prominence is and is not desirable.

3. Duopoly with exogenous utilities

As in the preceding example there is a prominent seller A and a follower seller B. A consumer with search cost $s \geq 0$ and return cost $r > 0$ has unit demand and her values for the products of the two sellers are uncertain and drawn independently from distribution F, with support $[\underline{u}, \bar{u}]$ (possibly infinite) and density f. For simplicity we assume as in Anderson and Renault (1999) that the consumer has no outside option and must choose from among A and B.

Define $w(z) \equiv \int_z^{\bar{u}} (u - z) f(u) du$ as the expected improvement over z from an additional draw. A consumer's continuation value from searching after initially observing u_A is

$$V(u_A) = -s + u_A - r + w(u_A - r),$$

in which after a sunk search cost of s, the consumer has locked in a return value of $u_A - r$ and has a potential improvement of $w(u_A - r)$. Because $V(u_A) - u_A$ is monotonically decreasing in u_A, the consumer's optimal policy is given by threshold $\hat{u}$ that satisfies

$$0 = -s - r + w(\hat{u} - r). \tag{1}$$

The probability the consumer buys from A is then

$$Q_A = \int_{\underline{u}}^{\hat{u}-r} (1 - F(u_B + r)) f(u_B)\, du_B \; + \; \int_{\hat{u}-r}^{\bar{u}} (1 - F(\hat{u})) f(u_B)\, du_B,$$

where the first term integrates over all sufficiently small u_B for which the consumer may return to A after searching, and the second term integrates over the larger values of u_B for which A makes the sale only if he initially deters search. Similarly, the probability the consumer buys from B is

$$Q_B = \int_{\underline{u}}^{\hat{u}} (1 - F(u_A - r)) f(u_A)\, du_A,$$

where if $u_A < \hat{u}$ then B sells as long as A's offer is not better by more than r, and if $u_A \geq \hat{u}$ then B does not sell at all.

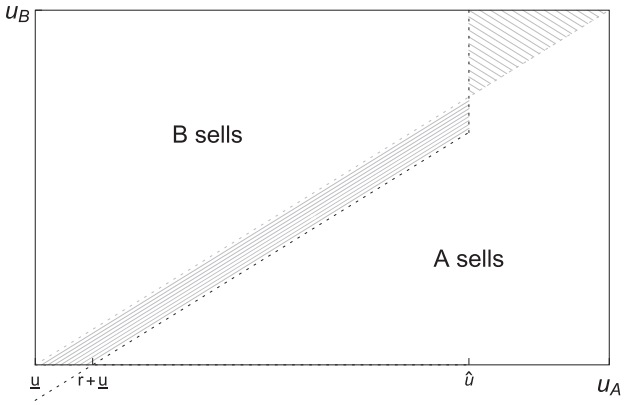


Fig. 1. Sales by leader A and follower B.

Let $\Delta Q \equiv \frac{1}{2}(Q_B - Q_A)$ represent one half of the net benefit to being visited second rather than first,[1] and observe that

$$\Delta Q(r,s) = \underbrace{\int_{\underline{u}}^{\underline{u}+r} F(u)f(u)\,du + \int_{\underline{u}+r}^{\hat{u}} \Big(F(u) - F(u-r)\Big)f(u)du}_{\text{benefit of later}} - \underbrace{\int_{\hat{u}}^{\bar{u}} (1-F(u))f(u)du}_{\text{benefit of prominence}}, \tag{2}$$

where r and s enter indirectly through search threshold $\hat{u}$ and r also enters directly in the first and second terms. Relative to the allocation with no frictions given by the diagonal forty-five degree line, A benefits from being first when search is pre-empted and consumers do not discover B's higher offer, as captured by the third term and depicted by the shaded triangle in Fig. 1, while B benefits from being last when return to A is pre-empted, as captured by the first two terms and the shaded strip near the diagonal in Fig. 1. The comparison of the two effects depends on the probability mass in the two shaded regions, and is not immediate because $\hat{u}$ is endogenously determined by the distribution F and the search and return costs. The marginal impact of a higher return cost is also ambiguous as there are countervailing effects – a higher r reduces the probability of return (larger second term) but also reduces the search threshold $\hat{u}$ (larger third term). We wish to establish the distributional properties by which to sign ΔQ and the comparative statics with respect to r.

Proposition 1. *ΔQ increases (decreases) in r whenever f is decreasing (increasing).*

[1] We define ΔQ as one half of the quantity difference to obtain cleaner expressions throughout the analysis. Because our interest will be in the signs of the quantity difference and its derivatives and not magnitudes, defining ΔQ this way has no impact.

This result coincides with our intuition for the tradeoffs associated with return costs. A higher r increases the follower's advantage for low outcomes and, by reducing search, increases the set of high outcomes for which the prominent seller has an advantage. The net effect depends on the relative likelihood of low versus high outcomes, which by Proposition 1 is captured by the slope of density f.

Proposition 1 is closely related to Proposition 2 in Armstrong and Zhou (2015), which compares a seller's payoff from imposing either no return costs or infinitely high return costs on visiting buyers, and finds the former (latter) is optimal when f is decreasing (increasing). The authors' quite elegant argument relies on Jensen's inequality to compare these two extreme options, but is not directly useful for our purposes because we must evaluate payoffs at an exogenous intermediate value for r. Our marginal result, that $\frac{d\Delta Q}{dr}$ is positive (negative) at every r when f is decreasing (increasing) in fact implies theirs. These comparative statics, combined with the implication of (1) and (2) that ΔQ strictly decreases in s, then yields the following result on when earlier or later is better.

Proposition 2. *If f is increasing then earlier is better $(\Delta Q \leq 0)$ for any $(r,\ s)$. If f is decreasing then later is better $(\Delta Q \geq 0)$ for any r and $s \in [0,\ s(r))$.*

In addition, we can identify whether prominence or later is better for small search and return costs even if density f is not monotonic, as follows.

Proposition 3. *Later is better for sufficiently small s and r if*

$$\int_{\underline{u}}^{\bar{u}} f^2(u)\ du > f(\bar{u}).$$

In particular, later is better if f is single-peaked and symmetric or if the support of f is right-unbounded.

It is instructive to consider the uniform distribution as an example. Because in this case f is constant throughout the support, the expression in the proposition holds with equality. Furthermore, from Proposition 2 it can be seen that for any r prominence is weakly better for $s = 0$ and strictly better for any $s > 0$. What makes the uniform distribution unappealing for being later rather than first? The answer is partly due to the term $\int_{\underline{u}}^{\bar{u}} f^2(u)du$, which measures the likelihood of exact ties for two random draws. When this likelihood is small, so too is the probability that the return cost sways a searcher to accept B rather than return to A, and the uniform specification *minimizes* this likelihood among all distributions with the same support. By contrast, a distribution with a single peak or a negative slope increases the odds of low and intermediate ties and makes being second relatively more attractive.[2]

[2] An upward sloping density, on the other hand, increases the odds of high ties, which is not helpful for seller B if these ties occur above the search threshold.

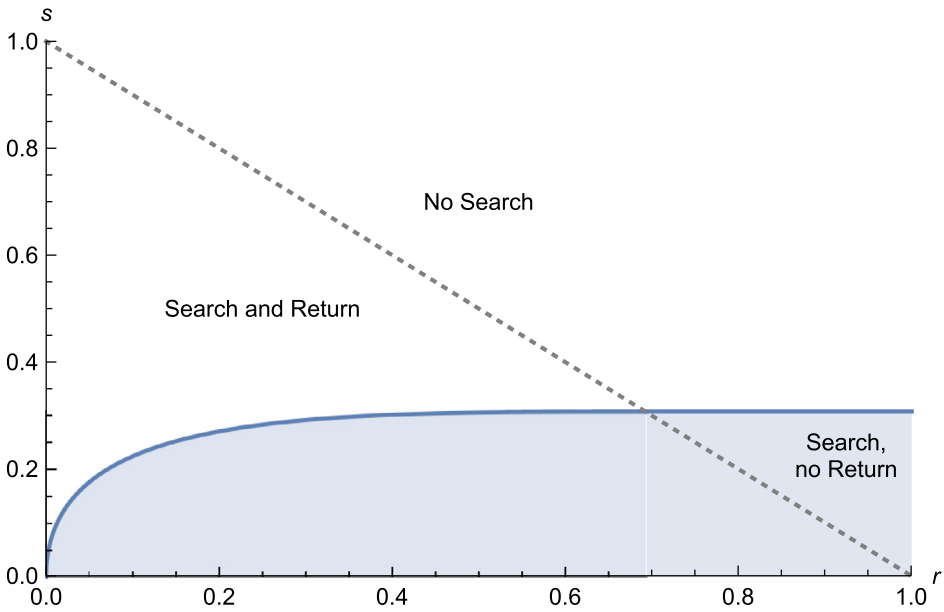


Fig. 2. Values of s and r for which later is better (shaded) for exponential distribution with $\lambda = 1$.

To demonstrate that later can be better for a reasonably large set of parameters, we consider as an example the exponential distribution $F(u) = 1 - e^{-\lambda u}$, with support $[0, \infty]$ and corresponding downward-sloping density $f(u) = \lambda e^{-\lambda u}$. Proposition 2 states that because f is downward-sloping, for every r there is a small enough s for which later is better, and indeed we can identify this set of parameters precisely. The consumer's utility threshold from (1) solves

$$s + r = \int_{\hat{u}-r}^{\infty} e^{-\lambda u}\, du \quad \Rightarrow \quad \hat{u} = r - \frac{1}{\lambda}\ln(\lambda(s+r)),$$

and can then be plugged into (2) to obtain

$$\Delta Q = \frac{1}{2}e^{-\lambda r}\left(e^{\lambda r} - (\lambda(s+r))^2 - 1\right).$$

Inspection of this expression reveals that it equals zero at $s = r = 0$, increases in r, and decreases in s. Therefore, for any fixed λ there is a frontier of search and recall costs given by $s = \frac{1}{\lambda}\left(e^{\lambda r} - 1\right)^{\frac{1}{2}} - r$ such that prominence is better whenever costs are above the frontier and later is better whenever costs are below the frontier.

Fig. 2 depicts this frontier for the case of $\lambda = 1$. The dotted line in the unshaded region corresponds to the largest search and return costs for which search can be supported, that is $\hat{u} \geq 0 \;\Rightarrow\; s + r \leq \frac{1}{\lambda}$. In the shaded region search always occurs, and consumers return with positive probability only if the return cost is to the left of the diagonal.

As the figure demonstrates, later is better for a substantial proportion of parameters in which any search occurs.

4. Duopoly with prices

We have shown that the prominent seller can be at a disadvantage when utilities are exogenous, but would such a result still hold if sellers could alter their offers prior to consumer search? To explore this question we adapt the differentiated product setting of Wolinsky (1986). The game proceeds in two stages: in the first stage the prominent seller A and follower B simultaneously set prices p_A and p_B, and in the second stage consumers search A and then potentially B as in the main model above. A consumer's utility for seller i's product is $u_i = \varepsilon_i - p_i$, where $\varepsilon_i \sim G$ with log-concave density g and support on $[\underline{\varepsilon}, \bar{\varepsilon}]$. The distribution G is commonly known but a consumer must visit seller i to learn both p_i and realization ε_i. Search cost s and return cost r apply as before.

Because match values are drawn independently consumers optimally follow the threshold strategy from (1), which when prices are included yields

$$0 = -s - r + w(\hat{u} + p_B - r). \tag{3}$$

Note that although only B's price affects $\hat{u}$, both prices affect the probability of search $\Pr(\varepsilon_A - p_A \geq \hat{u}) = 1 - G(\hat{u} + p_A)$. The quantities sold by A and B are now

$$Q_A = \int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} (1 - G(\varepsilon_B + p_A - p_B + r))g(\varepsilon_B)\, d\varepsilon_B + \int_{\hat{u}+p_B-r}^{\bar{\varepsilon}} (1 - G(\hat{u}+p_A))g(\varepsilon_B)d\varepsilon_B \tag{4}$$

and

$$Q_B = \int_{\underline{\varepsilon}}^{\hat{u}+p_A} (1 - G(\varepsilon_A - p_A + p_B - r))g(\varepsilon_A)\, d\varepsilon_A, \tag{5}$$

and each seller chooses his price to maximize revenue $\pi_i = p_i Q_i$.

We begin by showing that relative to the setting with exogenous utilities, the presence of an ex-ante pricing stage can favor the prominent seller over the follower.

Proposition 4. *If r is sufficiently small and (s, r) are such that earlier is better with exogenous utilities ($\Delta Q \leq 0$), then earlier is better in an equilibrium with prices ($\Delta\pi \leq 0$).*

The driving force behind this result is that the early seller faces a more elastic demand than the later seller. As in Armstrong et al. (2009), this is due to the fact that by raising his price the early seller can induce a consumer to search instead of buying immediately,

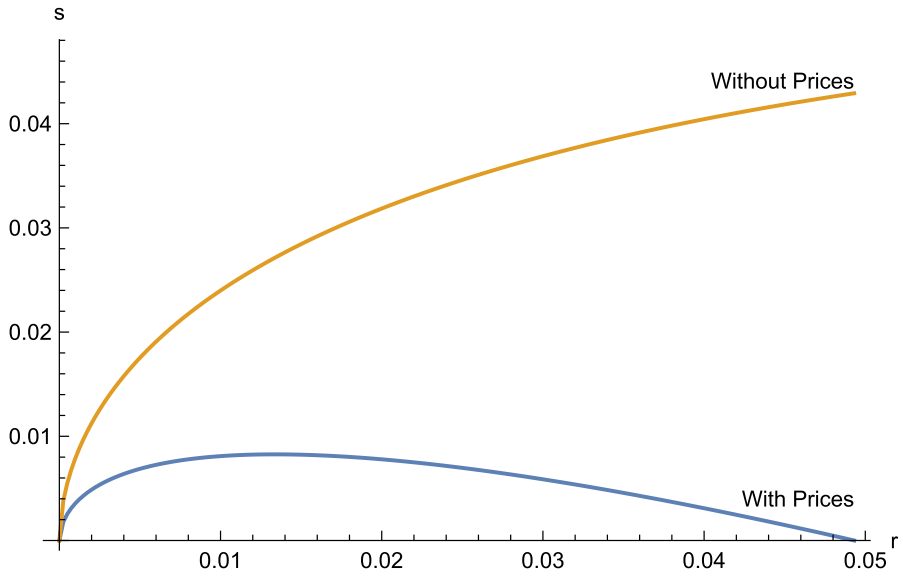


Fig. 3. Values of s and r below which later is better with and without prices for the standard normal distribution.

while the follower faces no such margin.[3] Consumers thus expect a higher price at the follower and are more reluctant to search, which favors the earlier seller.

To get a sense of how the first stage pricing game affects the value of search position we consider two match distributions as examples: the standard normal and the exponential. For the normal distribution, numerically solving for the equilibrium reveals that the earlier seller A charges a strictly lower price than the later seller B, and consequently the set of parameters for which later is better is smaller than in the setting without prices, as depicted in Fig. 3. However, for the exponential distribution, due to the fact that the hazard ratio is constant both sellers actually charge the same price in equilibrium,[4] and therefore later is better with prices exactly when later is better without prices, as in Fig. 2.

The first stage pricing game introduces a commitment problem for the later seller B: because he has an incentive to set a higher price than A, consumers expect this to happen and search less. Indeed, if B could commit to simply matching A's price, as happens endogenously with the exponential distribution, then B would be better off than A with prices whenever he is better off without prices.

Despite the fact that prominence can be more favorable when prices are endogenous, we can still derive a condition analogous to that in Proposition 3 to demonstrate that later can be better.

[3] Armstrong et al. (2009) show $p_A \leq p_B$ for a uniform density and $r = 0$, and here we extend this to any log-concave density and small values of r. We conjecture that the result holds for larger values of r as well, although we do not show this here formally.

[4] This is demonstrated in the Appendix.

Proposition 5. *If $g(\bar{\varepsilon}) = 0$ then later is better in an equilibrium with prices for sufficiently small s and r.*

As before, the return cost reduces the probability of search (due to anticipating costlier return and now also a higher p_B) and reduces the probability of return conditional on search. At $s = r = 0$ and threshold $\hat{\varepsilon} = \bar{\varepsilon}$, the former effect is of second order because there is very little probability mass around the search threshold, while the latter effect is of first order. Therefore, the increase in r induces an inward demand shift for A and an outward demand shift for B. When incorporated into the first stage pricing game, this generates lower profits for A and higher profits for B. See the Appendix for a detailed proof.

5. Oligopoly with exogenous utilities

We have demonstrated that in the case of two sellers later can be better whenever search is sufficiently likely and return sufficiently unlikely, specifically when high outcomes have low probability or when low and intermediate ties are relatively likely. We now investigate whether these positional tradeoffs apply in a setting with more than two sellers, and in particular when the optimal search position is first, last, or possibly somewhere in between. We begin with an example with three sellers and then present more general results.

Example with three sellers

A consumer can visit up to three sellers A, B, and C (in order), with utility at each seller drawn independently and given by

$$u_i = \begin{cases} v - \delta & \text{with probability} \quad x \\ v & \text{with probability} \quad 1 - 2x \\ v + \delta & \text{with probability} \quad x \end{cases}.$$

Assume $s \leq v$ so that the first search is optimal, and $s \leq \delta$ so that searching after receiving the lowest offer is optimal.

Lemma 1. *Define $\delta_1 \equiv \frac{x}{x+1} r + \frac{s}{x}$ and $\delta_2 \equiv r + \frac{s}{x}$. Then it is optimal for the consumer to use search thresholds $\hat{u}_i$ (search iff $u_i < \hat{u}_i$) at each of the first two sellers as follows:*

$$(\hat{u}_A, \hat{u}_B) = \begin{cases} (v, v) & \text{if} \quad \delta \in [0, \delta_1) \\ (v + \delta, v) & \text{if} \quad \delta \in [\delta_1, \delta_2) \\ (v + \delta, v + \delta) & \text{if} \quad \delta \in [\delta_2, \infty] \end{cases}.$$

Lemma 1 divides the parameter space (s, r, x, δ) into three regions, the high search region $\delta \in [\delta_2, \infty]$, the intermediate search region $\delta \in [\delta_1, \delta_2)$, and the low search region

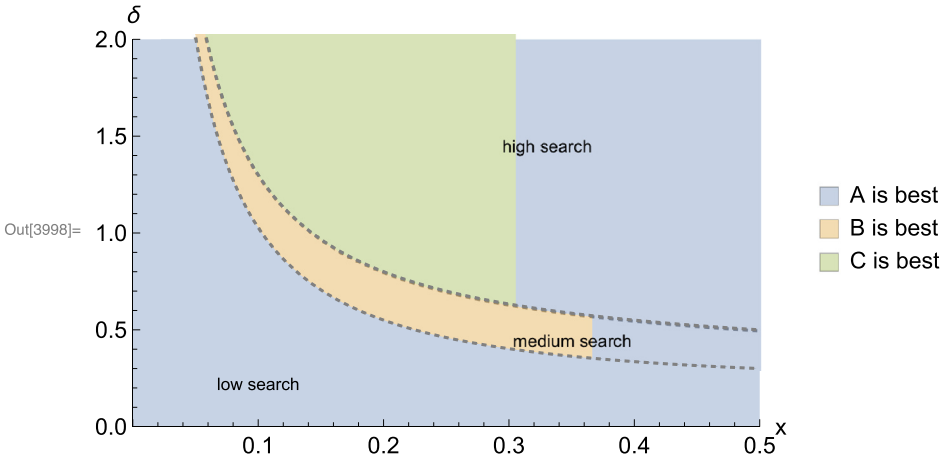


Fig. 4. Sales by A, B, and C, when $s = 0.3$ and $r = 0.1$.

$\delta \in [0, \delta_1)$. In the high search region the initial seller A wins ties over high outcomes while the last seller C wins ties over middle and low outcomes. In the intermediate second region the initial seller A wins ties over high outcomes, the second seller B wins ties over intermediate outcomes, and the final seller C wins ties over low outcomes. Finally, in the low search region the initial seller A wins ties over high and middle outcomes, while the last seller C wins ties over low outcomes. As we now show, depending on the region it is possible that either first, second, or third is best.

Proposition 6. *Let* $x_1 \equiv \frac{9-\sqrt{17}}{16} \approx 0.30$ *and* $x_2 \equiv \frac{\sqrt{3}-1}{2} \approx 0.37$. *Then,*

i. *if* $\delta \in [0, \delta_1)$ *then first is best,*
ii. *if* $\delta \in (\delta_1, \delta_2)$ *then second is best if* $x \in [0, x_2)$ *and first is best if* $x \in (x_2, 1/2]$,
iii. *if* $\delta \in (\delta_2, \infty)$ *then third is best if* $x \in [0, x_1)$ *and first is best if* $x \in (x_1, 1/2]$.

Fig. 4 depicts a set of distributional parameters x and δ for a fixed s and r, split into three regions depending on the intensity of search. In the southwest region the return to search is low and consequently A is best. In the middle region the return to search is intermediate, and in particular consumers are selective at A but not at B. Here B is best for lower values of x (i.e. when the middle outcome is likely) and A is best for higher values of x. Finally, in the northeast region in which the return to search is high it is best to be C for low values of x (i.e. when it is likely consumers make it through search and C benefits from the return cost) and best to be A for high values of x.

The example thus illustrates that the return cost bestows an advantage for all sellers, including intermediate sellers, for different realizations of match values. Depending on the distribution, it is possible that any place in the search order can be best.

Setting with many sellers

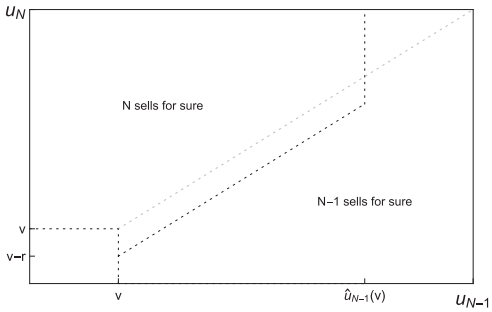
We now generalize some of the observations from the preceding example. To begin, note that in the absence of return costs the analysis is straightforward. Consumers would use a stationary search threshold $\hat{u}$ and because every earlier seller $i-1$ has an advantage over the next seller i whenever $\hat{u} \leq u_{i-1} < u_i$ and there is no return cost advantage, earlier is always better and first is best. However, with positive return costs this logic is complicated in two ways. First, as in the preceding analysis of two sellers, the final seller has a return cost advantage over each earlier seller whenever the consumer reaches the end. Second, the search threshold is in fact not stationary, as shown in Lemma 2 below, and instead decreases with the number of remaining observations. As the threshold falls from $\hat{u}_{i-1}$ to $\hat{u}_i$, the later seller gains an advantage over the earlier seller for realizations between the two thresholds. This can lead to first, last, or even in between to be the best position, as we discuss in more detail below.

Lemma 2. *(From Janssen and Parakhonyak, 2014) Define thresholds u^- and u^+ so that $s + r = w(u^- - r)$ and $s = w(u^+)$, and let v denote the best previously observed offer. Then, the optimal search strategy is characterized by threshold rule $\hat{u}_i(v) \in [u^-, u^+]$ which increases in v and decreases in i. When following this strategy the consumer returns only after having sampled every seller.*

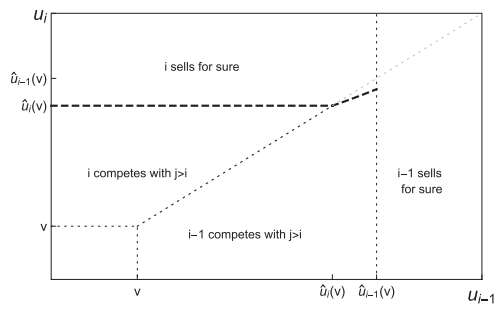
The reason that the threshold declines with fewer sellers remaining is that the likelihood of incurring the return cost increases, which makes search less attractive. The upper bound u^+ corresponds to infinitely many future draws and thus no chance of incurring the return cost while the lower bound u^- corresponds to a single future draw (as in (1)) and thus, the maximum probability of incurring the return cost.

To describe the value of a particular search position it is useful to establish the marginal benefit of moving one spot in the search order, and to do so we make two types of comparisons: the penultimate seller $N-1$ with the last seller N, and any earlier seller $i-1$ with the next seller $i < N$. First, having arrived at the penultimate seller $N-1$ with a previous best offer of v, a consumer accepts if $u_{N-1} \geq \hat{u}_{N-1}(v)$, else she searches and buys from N if $u_N \geq \max(u_{N-1}, v) - r$, returns to $N-1$ if $u_{N-1} \geq \max(v, u_N + r)$, and returns to the previous best offer v otherwise, as depicted in the left panel of Fig. 5. The tradeoff is thus the same as in the case of two sellers except for the presence of the previous best offer v, which has countervailing effects. It helps the last seller by increasing the search threshold $\hat{u}_{N-1}$ but also reduces the last seller's return cost advantage, as u_N must exceed both $u_{N-1} - r$ and $v - r$ in order for N to make the sale. The net effect is potentially ambiguous.

Second, the tradeoff between middle sellers $i-1$ and i is affected by the fact that consumers become less selective as search continues, and is depicted in the right panel of Fig. 5. A consumer visiting $i-1$ with a best previous offer of v buys immediately if $u_{i-1} \geq \hat{u}_{i-1}(v)$, else she visits i and then buys immediately if $u_i \geq \hat{u}_i(\max(u_{i-1}, v))$, given



(a) Sales by final sellers $N-1$ and N



(b) Sales by interior sellers $i-1$ and i

Fig. 5. Sales to a consumer with a best previous offer v arriving at penultimate seller $N-1$ (left panel) and intermediate seller $i-1$ (right panel).

by the dark dashed line in the figure, and otherwise she keeps searching and potentially returns to either seller only upon reaching the end. Seller i is thus less likely to be reached than seller $i-1$, but conditional on being reached has a bigger search advantage than $i-1$ because $\hat{u}_{i-1}(v) > \hat{u}_i(\max(u_{i-1}, v))$, as stated in Lemma 2. Again, the net effect is potentially ambiguous. With the marginal positional effects thus described, we can establish some conditions under which first or last is best.

Proposition 7. *If $f(\bar{u}) = 0$ then for any number of sellers N, last is best for sufficiently small s and r.*

This result is analogous to Proposition 3 for the case of two sellers. As demonstrated in the proof in the Appendix, this follows because at $s = r = 0$ an increase in r has no first order effect on the tradeoff between i and $i-1$ and favors seller N over seller $N-1$. Put differently, last can be best even when N is quite large as long as frictions are small enough so that the last seller is likely to be reached.

On the other hand, if search and return costs are fixed and more sellers are added, this not only harms the final seller by reducing the probability of being reached but also aids the earliest seller relative to the already existing sellers that followed him.

Proposition 8. *For any fixed $s \geq 0$ and $r > 0$ there exists a large enough N so that first is best.*

The intuition for this result comes from the fact that the threshold is essentially stationary when a large number of samples remains, and thus the size of the interval $[\hat{u}_i(v), \hat{u}_{i-1}(v)]$ over which a later seller i derives an advantage over her predecessor $i-1$ shrinks toward zero. Consequently earlier is better early on, and if this tradeoff flips at some later seller, it happens so late that this seller is unlikely to ever be reached.

Proposition 8 is consistent with recent empirical evidence (Novarese and Wilson, 2013; Ho and Imai, 2008) of list length increasing the advantage of the top position. Such an

effect is difficult to rationalize without appealing to specific behavioral biases, as having more options available later should make a buyer more selective, not less. Our framework is the first to our knowledge to provide a rational explanation.

Propositions 7 and 8 demonstrate that first or last can be best when parameters are taken toward their limits. A more general treatment of when first, last, or perhaps even somewhere in the middle is best is difficult because of complexity associated with the non-stationarity of the search threshold.[5] However, the preceding example does demonstrate that with three sellers first, second, or third can each be best.

6. Conclusion

While it is commonly held that appearing earlier in a consumer's search is desirable for a seller, we demonstrate that this need not be the case. By allowing for a more realistic environment in which it is both costly to obtain new offers and to recall previous offers, we show that it is sometimes better to be later in the search order. Specifically, the search and return frictions endow the first position with an advantage over high match outcomes and the last position with an advantage over low match outcomes. In addition, because consumers become less selective during search, middle positions get an advantage over intermediate outcomes. Which position is optimal then depends on which range of outcomes has the highest probability.

Our analysis applies to several settings in which search order plays an important role. For example, if a platform sells top search positions then the characteristics of the match value distribution for different types of content help determine the demand for top spots. Pricing for prominence in search results should depend not only on the volume of traffic for the content, but also on the properties of the match value distributions that we describe. These considerations can guide the platform not only in the sale of prominent spots but also in the platform's design and organization. For instance, allowing for longer, more detailed blurbs provides a more informative signal for users, thereby leading to a distribution of match values where ties are less likely, and thus a higher return to prominence, while shortening or eliminating blurbs all together has the opposite effect.

Alternatively, consider a labor market in which an employer successively interviews job candidates. Other things being equal, how much more likely is a candidate to be hired if she is interviewed first and thus how much effort should she expend to be among the first interviewees? Or should she, to the contrary, strive to be interviewed last? If the distribution of candidates in the pool is such that most candidates fit the position similarly but with a small chance the match is outstanding, an employee is more likely to be hired if she is interviewed late rather than early. Here too the employer's design of the interview process affects the probability that earlier interviewees are hired. For example, requiring

[5] In particular, computing whether position $i-1$ or $i<N$ is better depends on the best previous offer v, and thus one must integrate over all possible histories to make the comparison. Furthermore, for each such history one must keep track of the evolving search threshold, and of whether it is ever exceeded prior to arriving at $i-1$.

more or less detailed pre- interview information and documentation from applicants affects the distribution of match values between the employer and job applicants, which in turn determines the extent to which earlier applicants are more likely to be hired.

Our model may also be thought of as the second stage of a game in a market in which firms compete for prominence in the first stage. Sellers may compete for prominence through advertising (Haan and Moraga-González, 2011), price setting (Choi et al., 2016; Haan et al., 2017; Armstrong and Zhou, 2011), obfuscation or disclosure (Wilson, 2010), or other methods. Our results suggest that the intensity of the first stage competition strongly depends on the magnitudes of the search and return costs and that, in particular, return costs may mitigate or even eliminate first stage competition for prominence.

Appendix

A.1. Proof of Proposition 1

Differentiating (2) and simplifying yields

$$\frac{d(\Delta Q)}{dr} = \int_{\underline{u}+r}^{\hat{u}} f(u-r)f(u)du \quad + \quad f(\hat{u})(1-F(\hat{u}-r))\frac{d\hat{u}}{dr}.$$

Then totally differentiating (1) yields $\frac{d\hat{u}}{dr} = -\frac{F(\hat{u}-r)}{1-F(\hat{u}-r)}$, and plugging into the above obtains

$$\frac{d(\Delta Q)}{dr} = \int_{\underline{u}+r}^{\hat{u}} f(u-r)f(u)du \quad - \quad f(\hat{u})F(\hat{u}-r). \tag{6}$$

If f is decreasing then

$$\begin{aligned}\frac{d(\Delta Q)}{dr} &= f(\hat{u})\left(\int_{\underline{u}+r}^{\hat{u}} f(u-r)\frac{f(u)}{f(\hat{u})}\,du \;-\; F(\hat{u}-r)\right)\\ &\geq f(\hat{u})\left(\int_{\underline{u}+r}^{\hat{u}} f(u-r)\,du \;-\; F(\hat{u}-r)\right)\\ &= 0.\end{aligned}$$

By the same logic, if f is increasing then $\frac{d(\Delta Q)}{dr}$.

A.2. Proof of Proposition 2

If f is increasing then $0 = \Delta Q(0,0) > \Delta Q(r,0) > \Delta Q(r,s)$, where the first inequality follows from Proposition 1 and the second inequality follows from ΔQ decreasing in s. If f is decreasing then $0 < \Delta Q(r,0) \leq \Delta Q(r,\hat{s})$, where the first inequality again follows from Proposition 1 and the existence of a small enough $\hat{s}$ to satisfy the second inequality comes from the continuity of ΔQ in s.

A.3. Proof of Proposition 3

Evaluating (6) at $r = 0$ and $s = 0$ and corresponding threshold $\hat{u} = \bar{u}$ yields

$$\frac{d(\Delta Q)}{dr} = \int_{\underline{u}}^{\bar{u}} f^2(u)du \quad - \quad f(\bar{u}). \tag{7}$$

The right-hand side of (7) is positive by assumption, and therefore ΔQ increases in r at $s = r = 0$. Then by continuity ΔQ is positive in a neighborhood of $s = r = 0$. Observe that the right hand side of (7) is the difference between the average density and the density at the upper bound, and that this expression is positive for symmetric single-peaked and for right-unbounded distributions because in both cases $f(\bar{u})$ is the smallest density in the entire support.

A.4. Proof that $p_A = p_B = \frac{1}{\lambda}$ with exponential match values

From each seller's first order condition the optimal prices is given by $p_i = -\frac{Q_i}{\frac{\partial Q_i}{\partial p_i}}$. For seller B, using (5) this expression is

$$p_B = \frac{\int_{\underline{\varepsilon}}^{\hat{u}+p_A} e^{-\lambda(2\varepsilon_A - p_A + p_B - r)}\, d\varepsilon_A}{\int_{\underline{\varepsilon}}^{\hat{u}+p_A} \lambda e^{-\lambda(2\varepsilon_A - p_A + p_B - r)}\, d\varepsilon_A} = \frac{1}{\lambda}.$$

For A, using (4) this expression is

$$p_A = \frac{\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} e^{-\lambda(2\varepsilon_B + p_A - p_B + r)}\, d\varepsilon_B \;+\; e^{-\lambda(2\hat{u}+p_B+p_A-r)}}{\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} \lambda e^{-\lambda(2\varepsilon_B + p_A - p_B + r)}\, d\varepsilon_B \;+\; \lambda e^{-\lambda(2\hat{u}+p_B+p_A-r)}} = \frac{1}{\lambda}.$$

A.5. Proof of Proposition 4

We first establish that $p_A \leq p_B$ for small values of r. Then we use this fact demonstrate the statement of the proposition, that for small r if earlier is better with exogenous utilities then earlier is also better in an equilibrium with prices.

Claim 1. *For a weakly increasing $k(\cdot)$, the function $\varphi(b) \equiv \frac{\int_{\underline{\varepsilon}}^{b} g(k(\varepsilon))g(\varepsilon)d\varepsilon}{\int_{\underline{\varepsilon}}^{b}(1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon}$ increases in b.*

Proof of Claim Observe that

$$\frac{\partial \varphi}{\partial b} = \frac{(1 - G(k(b)))g(b)}{\left(\int_{\underline{\varepsilon}}^{b}(1 - G(k(\varepsilon)))g(\varepsilon)d\varepsilon\right)^2}$$

$$\times \left(\frac{g(k(b))}{1-G(k(b))} \int_{\underline{\varepsilon}}^{b} (1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon - \int_{\underline{\varepsilon}}^{b} \frac{g(k(\varepsilon))}{1-G(k(\varepsilon))}(1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon \right)$$

$$\geq \frac{(1-G(k(b)))g(b)}{\left(\int_{\underline{\varepsilon}}^{b}(1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon\right)^2}$$

$$\times \left(\frac{g(k(b))}{1-G(k(b))} \int_{\underline{\varepsilon}}^{b} (1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon - \int_{\underline{\varepsilon}}^{b} \frac{g(k(b))}{1-G(k(b))}(1-G(k(\varepsilon)))g(\varepsilon)d\varepsilon \right)$$

$$= 0,$$

where the inequality follows from the fact that the hazard ratio $\frac{g(\cdot)}{1-G(\cdot)}$ is increasing (due to the log-concavity of g), and therefore $\frac{g(k(b))}{1-G(k(b))} \geq \frac{g(k(\varepsilon))}{1-G(k(\varepsilon))}$.

Claim 2. *If r is sufficiently small then in equilibrium $p_A \leq p_B$.*

Proof of Claim. Define $\hat{\varepsilon}$ as the solution to $s + r = w(\hat{\varepsilon} - r)$, $k(x) = x$, and corresponding function φ as in Claim 1. Given $r = 0$ and both sellers charge the same price p, it follows that

$$\frac{|dQ_A/dp_A|}{Q_A} \geq \varphi(\hat{\varepsilon}) = \frac{|dQ_B/dp_B|}{Q_B},$$

where the inequality follows by the preceding claim and the equality by definition. Thus when charging the same price A's demand is more elastic than B's when $r = 0$, and by continuity also in the neighborhood of $r = 0$. Consequently, in this neighborhood $p_A \leq p_B$. □

Now we can prove the main statement of the proposition. Conjecture an equilibrium with prices $(p_A,\ p_B)$ and observe that if seller A deviates to match B's price then his profit would be higher than B's, i.e. $\pi_A(p_B, p_B) = p_B Q_A(p_B, p_B) \geq p_B Q_B(p_B, p_B)$, with the inequality following from the assumption that $\Delta Q \leq 0$. But then

$$\pi_A(p_A, p_B) \geq \pi_A(p_B, p_B) \geq \pi_B(p_B, p_B) \geq \pi_B(p_B, p_A),$$

where the final inequality follows because $p_A \leq p_B$ and B's profit falls as p_A falls. Observe that A's price does not affect the search threshold $\hat{u}$, thus it is omitted from the arguments of π.

A.6. Proof of Proposition 5

As in the case without prices, we argue that $\Delta\pi \equiv \pi_B - \pi_A$ increases in r at $s = r = 0$, and therefore there is an $(s,\ r)$ neighborhood in which later is better. We demonstrate that it is sufficient to track $\frac{d\Delta Q}{dr}$ by appealing to the envelope theorem, but now we must account for the fact that r changes ΔQ both directly and through prices. We first

establish in Claim 3 that in a neighborhood of $s = r = 0$ there exists a pure strategy pricing equilibrium. Then we show in Claim 4 that $\frac{d(p_B - p_A)}{dr} < 1$, so that the increase in the price gap does not exceed the return advantage, thus on net an increase in r still reduces the probability of return conditional on search. Finally, we demonstrate that as before the reduction in search due to a higher r is of second order because by assumption there is no probability weight near the high search threshold.

Claim 3. *There exists a pure strategy equilibrium in prices in the neighborhood of* $s = r = 0$.

Proof of Claim. At $s = r = 0$ we are in the setting of Perloff and Salop (1985), where the log-concavity of g implies the log-concavity of Q_A and Q_B by the Prékopa–Borell theorem, and in turn A and B's payoffs are log-concave in own price. Then, by continuity A and B's payoffs are quasi-concave (which is weaker than log-concave) in the neighborhood of $s = r = 0$, and thus by standard fixed point arguments a pure strategy equilibrium exists. □

Claim 4. $\frac{d(p_B - p_A)}{dr} \leq 1$ *at* $s = r = 0$.

Proof of Claim. Let $\Delta \equiv r + p_A - p_B$. As previously, define $\varphi_i \equiv \left|\frac{\partial Q_i}{\partial p_i}\right| / Q_i$. Starting with seller A,

$$\varphi_A = \frac{\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} g(\varepsilon+\Delta)g(\varepsilon)d\varepsilon \;+\; \int_{\hat{u}+p_B-r}^{\bar{\varepsilon}} g(\hat{u}+p_A)g(\varepsilon)d\varepsilon}{\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r}(1-G(\varepsilon+\Delta))g(\varepsilon)d\varepsilon \;+\; \int_{\hat{u}+p_B-r}^{\bar{\varepsilon}}(1-G(\hat{u}+p_A))g(\varepsilon)d\varepsilon}$$

and the derivative with respect to r is

$$\frac{d\varphi_A}{dr} = \frac{1}{Q_A^2}\left[\left(\frac{d\Delta}{dr}\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} g'(\varepsilon+\Delta)g(\varepsilon)d\varepsilon + \frac{d(\hat{u}+p_A)}{dr}\int_{\hat{u}+p_B-r}^{\bar{\varepsilon}} g'(\hat{u}+p_A)g(\varepsilon)d\varepsilon\right)Q_A \right.$$
$$\left. + \left(\frac{d\Delta}{dr}\int_{\underline{\varepsilon}}^{\hat{u}+p_B-r} g(\varepsilon+\Delta)g(\varepsilon)d\varepsilon + \frac{d(\hat{u}+p_A)}{dr}\int_{\hat{u}+p_B-r}^{\bar{\varepsilon}} g(\hat{u}+p_A)g(\varepsilon)d\varepsilon\right)\left|\frac{dQ_A}{dp_A}\right|\right].$$

Evaluating at $s = r = 0$, we know that $\hat{u} + p_B - r = \bar{\varepsilon}$, $\Delta = 0$, $Q_A = Q_B \equiv Q$, and $\left|\frac{dQ_A}{dp_A}\right| = \left|\frac{dQ_B}{dp_B}\right| \equiv Q'$, the above expression simplifies to

$$\frac{d\varphi_A}{dr}(s = r = 0) = \frac{d\Delta}{dr}\left(\frac{\int_{\underline{\varepsilon}}^{\bar{\varepsilon}}(g'(\varepsilon)Q + g(\varepsilon)Q')g(\varepsilon)d\varepsilon}{Q^2}\right) = \frac{d\Delta}{dr}\varphi', \tag{8}$$

where in the last equality $\varphi' \equiv \frac{d}{d\Delta}|_{\Delta=0} \frac{\int_{\underline{\varepsilon}}^{\bar{\varepsilon}} g(\varepsilon+\Delta)g(\varepsilon)d\varepsilon}{\int_{\underline{\varepsilon}}^{\bar{\varepsilon}}(1-G(\varepsilon+\Delta))g(\varepsilon)d\varepsilon}$. Note that $\varphi' \geq 0$ by the log-concavity of $1-G$. Now performing the same analysis for B,

$$\varphi_B = \frac{\int_{\underline{\varepsilon}}^{\hat{u}+p_A} g(\varepsilon-\Delta)g(\varepsilon)d\varepsilon}{\int_{\underline{\varepsilon}}^{\hat{u}+p_A}(1-G(\varepsilon-\Delta))g(\varepsilon)d\varepsilon}$$

and the derivative with respect to r is

$$\frac{d\varphi_B}{dr} = \frac{1}{Q_B^2}\Bigg[-\frac{d\Delta}{dr}\left(\int_{\underline{\varepsilon}}^{\hat{u}+p_A} g'(\varepsilon-\Delta)g(\varepsilon)d\varepsilon + \frac{d(\hat{u}+p_A)}{dr}g(\hat{u}+p_B-r)g(\hat{u}+p_A)\right)Q_B$$
$$-\frac{d\Delta}{dr}\left(\int_{\underline{\varepsilon}}^{\hat{u}+p_A} g(\varepsilon-\Delta)g(\varepsilon)d\varepsilon + \frac{d(\hat{u}+p_A)}{dr}(1-G(\hat{u}+p_B-r))g(\hat{u}+p_A)\right)\left|\frac{dQ_B}{dp_B}\right|\Bigg].$$

Evaluating as before at $s=r=0$ and now also using the fact that $g(\bar{\varepsilon})=0$ yields

$$\frac{d\varphi_B}{dr}(s=r=0) = -\frac{d\Delta}{dr}\varphi'. \tag{9}$$

Putting (8) and (9) together yields

$$\frac{d\varphi_A}{dr} = \frac{d\Delta}{dr}\varphi' = -\frac{d\varphi_B}{dr}.$$

Now recall that each seller's optimal price satisfies $p_i = \frac{1}{\varphi_i}$ and suppose toward a contradiction that $\frac{d(p_B-p_A)}{dr} > 1$, so that $\frac{d\Delta}{dr} < 0$. Then $\frac{d\varphi_A}{dr} < 0$ and so $\frac{dp_A}{dr} > 0$, while $\frac{d\varphi_B}{dr} > 0$ and so $\frac{dp_B}{dr} < 0$. But this leads to a contradiction because $\frac{d(p_B-p_A)}{dr} < 0$ but by assumption $\frac{d(p_B-p_A)}{dr} > 1$. This concludes the proof of the claim. □

We can now prove the main statement of the proposition. Observe that equilibrium profits can be written as $\pi_i(p_i(r),r)) = p_i Q_i(p_i,p_j,\hat{u},r)$, and that therefore

$$\frac{d\pi_i}{dr} = \frac{\partial\pi_i}{\partial p_i}\frac{dp_i}{dr} + \frac{\partial\pi_i}{\partial r} = p_i\left(\frac{\partial Q_i}{\partial p_j}\frac{dp_j}{dr} + \frac{\partial Q_i}{\partial\hat{u}}\frac{d\hat{u}}{dr} + \frac{\partial Q_i}{\partial r}\right),$$

where the second equality follows by the envelope theorem. Then, given that $\pi_B = p_B Q_B$ and $\pi_A = p_A(1-Q_B)$,

$$\frac{d(\pi_B-\pi_A)}{dr} = p_B\left(\frac{\partial Q_B}{\partial p_A}\frac{dp_A}{\partial r} + \frac{\partial Q_B}{\partial\hat{u}}\frac{d\hat{u}}{dr} + \frac{\partial Q_B}{\partial r}\right) + p_A\left(\frac{\partial Q_B}{\partial p_B}\frac{dp_B}{dr} + \frac{\partial Q_B}{\partial\hat{u}}\frac{d\hat{u}}{dr} + \frac{\partial Q_B}{\partial r}\right).$$

Evaluated at $s=r=0$, we know that $p_A = p_B = p$ and $\frac{\partial Q_B}{\partial p_B} = -\frac{\partial Q_B}{\partial p_A} = -\frac{\partial Q_B}{\partial r} = Q'$, and then can rewrite the above difference as

$$\frac{d(\pi_B-\pi_A)}{dr}(s=r=0) = p\left(|Q'|\left(2-\frac{d(p_B-p_A)}{dr}\right) + 2\frac{\partial Q_B}{\partial\hat{u}}\frac{d\hat{u}}{dr}\right). \tag{10}$$

Recalling that

$$\frac{\partial Q_B}{\partial \hat{u}}\frac{d\hat{u}}{dr} = (1-G(\hat{u}+p_B-r))g(\hat{u}+p_A)\left(-\frac{G(\hat{u}+p_B-r)}{1-G(\hat{u}+p_B-r)}-\frac{dp_B}{dr}\right)$$
$$= -g(\hat{u}+p_A)G(\hat{u}+p_B-r)-(1-G(\hat{u}+p_B-r))g(\hat{u}+p_A)\frac{dp_B}{dr},$$

and that at $s=r=0$, $\hat{u}+p_A=\hat{u}+p_B+r=\bar{\varepsilon}$, we thus know that

$$\frac{\partial Q_B}{\partial \hat{u}}\frac{d\hat{u}}{dr}(s=r=0)=0.$$

We also know by Claim 4 that $\frac{d(p_B-p_A)}{dr}<1$, and therefore applying this to (10) yields $\frac{d(\pi_B-\pi_A)}{dr}(s=r=0)>0$. □

A.7. Definitions and proofs for Section 5: many sellers

Because the setting with N sellers introduces a layer of complexity, we first explicitly describe objects useful for the proofs of Propositions 7 and 8, and then present the proofs themselves.

A7.1. Definitions

Given the optimal search policy $\hat{u}_i(v)$ in Lemma 2, let $v_i \equiv \max_{k\le i} u_k$ be the best offer from the first i observations and define

$$\rho_i(u) \equiv \int_{u_{i+1}=\underline{u}}^{\min(u,\hat{u}_{i+1}(v_i))}\int_{u_{i+2}=\underline{u}}^{\min(u,\hat{u}_{i+2}(v_{i+1}))}\cdots\int_{u_{N-1}=\underline{u}}^{\min(u,\hat{u}_{N-1}(v_{N-2}))}\int_{u_N=\underline{u}}^{u-r}\left(\prod_{k=i+1}^{N} f(u_k)\,du_k\right)$$

as the probability that a consumer that observes u at seller i and keeps searching eventually chooses i over all later options. This involves future observations $u_{k<N}$ not exceeding either the current value u or their respective search thresholds $\hat{u}_k(v_{k-1})$, and the final observation u_N not exceeding $u-r$. For seller $i-1$, conditional on consumers arriving with a best offer v her probability of sale is given by

$$Q_{i-1}(v) = 1-F(\hat{u}_{i-1}(v)) \;+\; \int_{\hat{u}_i(v)}^{\hat{u}_{i-1}(v)} F(\hat{u}_i(u))\rho_i(u)f(u)du \;+\; \int_{v}^{\hat{u}_i(v)} F(u)\rho_i(u)f(u)du.$$

In the first term, for outcomes above $\hat{u}_{i-1}$ the consumer buys immediately. Otherwise the consumer only returns if she searches to the end and the best return option is u_{i-1}. With respect to later sellers $j>i$, seller $i-1$ wins with probability $\rho_i(u_{i-1})$, while with respect to seller i in particular, $i-1$ wins with probability $F(u_{i-1})$ if u_{i-1} is below i's search threshold (third term) and with probability $F(\hat{u}_i(u_{i-1}))$ if u_{i-1} is above i's search threshold (second term). We can similarly write the probability that i makes the sale

conditional on the consumer arriving at $i-1$ with best previous offer v as

$$Q_i(v) = (1 - F(\hat{u}_{i-1}(v)))F(\hat{u}_{i-1}(v)) + \int_{\hat{u}_i(v)}^{\hat{u}_{i-1}(v)} F(\min(\hat{u}_{i-1}(v), \hat{u}_i^{-1}(u)))f(u)du \; + \; \int_{v}^{\hat{u}_i(v)} F(u)\rho_i(u)f(u)du.$$

For high outcomes $u_i \geq \hat{u}_{i-1}$ in the first term, the consumer buys right away from i given she has not stopped searching at $i-1$. For intermediate outcomes $u_i \in (\hat{u}_{i-1}(v), \hat{u}_i(v))$ in the second term, the consumer also immediately buys from i given she has not stopped searching at $i-1$, which implies that $u_i \geq \hat{u}_i(u_{i-1}) \;\Leftrightarrow\; u_{i-1} \leq \hat{u}_i^{-1}(u_i)$. For low outcomes $u_i \in (v, \hat{u}_i(v))$ in the third term, the consumer searches and returns only if i beats all later sellers with probability $\rho_i(u_i)$ and the preceding seller with probability $F(u_i)$.

Taking the difference between the two quantities and simplifying yields

$$\Delta Q_i(v) \equiv Q_i(v) - Q_{i-1}(v) = \int_{\hat{u}_i(v)}^{\hat{u}_{i-1}(v)} \Big(F(\min(\hat{u}_{i-1}, \hat{u}_i^{-1}(u))) - F(\hat{u}_i(u))\rho_i(u)\Big)f(u)du - \frac{1}{2}(1 - F(\hat{u}_{i-1}(v)))^2. \tag{11}$$

The first term reflects the advantage to the later seller, due to the fact that for realizations between the earlier and later thresholds seller i makes the sale for sure when she is reached, while the earlier seller $i-1$ must compete both with the later seller i and all the sellers that follow. The second term then reflects the standard advantage of the earlier seller, making the sale with certainty when her search threshold is exceeded.

Expression (11) describes the payoff from moving down one spot in the search order for all positions except the last, where the tradeoff is instead given by

$$\Delta Q_N(v) \equiv \frac{1}{2}(Q_N(v) - Q_{N-1}(v)) = \int_{v}^{\hat{u}_{N-1}(v)} (F(u) - F(u-r))f(u)du - \frac{1}{2}(1 - F(\hat{u}_{N-1}(v)))^2. \tag{12}$$

This tradeoff differs from that of the two-seller case in (2) only in that the consumer arrives at $N-1$ with a best previous offer of v, and thus the threshold is now given by

$$\hat{u}_{N-1}(v) = \begin{cases} u^- & \text{if} \quad v \leq u^- \\ -s + v - r + w(v-r) & \text{if} \quad v \geq u^- \end{cases},$$

matching the $N=2$ threshold from Eq. (1) if $v \leq u^-$ and exceeding it if $v \geq u^-$.

A7.2. Proof of Proposition 7

We show that as r is increased at $s = r = 0$, at which point all sellers start on equal footing, there is no first order effect on the difference between interior sellers and a first order effect in favor of the last seller relative to the penultimate seller.

Beginning with interior sellers, measuring the change in the search thresholds with respect to r is quite involved, however instead of calculating this change precisely we use a bounds approach. To reduce notation we suppress the best previous offer v as an argument of quantities and thresholds, and define an upper bound

$$\Delta\tilde{Q}_i \equiv \int_{\hat{u}_i}^{\hat{u}_{i-1}} (F(u)(1-\rho_i(u))f(u)du - \frac{1}{2}(1-F(\hat{u}_{i-1}))^2.$$

This bound $\Delta\tilde{Q}_i$ differs from the actual quantity difference ΔQ_i in (11) for realizations in the interval $[\hat{u}_i, \hat{u}_{i-1}]$. In particular, it presumes that consumers use u_{i-1} as a search threshold at seller i, while in reality the threshold $\hat{u}_i(u_{i-1})$ is lower. This reduces the later seller i's benefit, and thus $\Delta\tilde{Q}_i \leq \Delta Q_i$. Then,

$$\begin{aligned}\frac{d\Delta\tilde{Q}_i}{dr} &= F(\hat{u}_{i-1})(1-\rho_i(\hat{u}_{i-1}))f(\hat{u}_{i-1})\frac{d\hat{u}_{i-1}}{dr} \;-\; F(\hat{u}_i)(1-\rho_i(\hat{u}_i))f(\hat{u}_i)\frac{d\hat{u}_i}{dr} \\ &\quad - \int_{\hat{u}_i}^{\hat{u}_{i-1}} F(u)f(u)\frac{d\rho_i}{dr}du \;-\; f(\hat{u}_{i-1})(1-F(\hat{u}_{i-1}))\frac{d\hat{u}_{i-1}}{dr}.\end{aligned}$$

Now observe that at $r = s = 0$ the thresholds all evaluate to $\bar{u}$ and the probability $\rho_i(\bar{u})$ evaluates to one. By inspection, all four of the above terms equal zero, and thus $\frac{d\Delta\tilde{Q}_i}{dr} = 0$ at $s = r = 0$. Because $\Delta Q_i = 0$ at $s = r = 0$, and $\Delta Q_i \geq \Delta\tilde{Q}_i$ at all s and r, it must be that $\frac{d\Delta Q_i}{dr} \geq 0$ at $s = r = 0$.

Next, turning to the tradeoff between the penultimate and final seller from (12) obtains

$$\frac{d\Delta Q_N}{dr} = \int_v^{\hat{u}_{N-1}} f(u-r)f(u)du \;-\; (1-F(\hat{u}_{N-1}-r))f(\hat{u}_{N-1})\frac{d\hat{u}_{N-1}}{dr}.$$

Again, when evaluated at $s = r = 0$ the threshold is $\hat{u}_{N-1} = \bar{u}$ and therefore the derivative simplifies to

$$\frac{d\Delta Q_N}{dr} = \int_v^{\bar{u}} f(u-r)f(u)du,$$

which is greater than zero whenever the best previous observation $v < \bar{u}$, which happens with positive probability. Thus, at $s = r = 0$ we obtain $E_v[\Delta Q_N] > 0$.

Putting the results together, because at $s = r = 0$ we obtain $\frac{d\Delta Q_i}{dr} \geq 0$ and $E_v\left[\frac{d\Delta Q_N}{dr}\right] > 0$, for sufficiently small s and r last is best.

A7.3. Proof of Proposition 8

The argument rests of the fact that $\lim_{N\to\infty} \hat{u}_i(v) = u^+$ as shown by Janssen and Parakhonyak (2014). Recalling Eq. (12), because $\hat{u}_{i-1} - \hat{u}_i$ can be made arbitrarily small, for any $\varepsilon > 0$ and any i and $M > i$, there exists an N large enough so that $\Delta Q_j(v) < \frac{1}{2}(1-F(u^+))^2 + \varepsilon$ for all $i \leq j \leq M$. In words, for any two sellers if N is large

enough then quantity falls monotonically from the earlier to the later seller. If M is the first seller at which $\Delta Q_M > 0$ then whenever M is large

$$Q_1 \geq 1 - F(u^+) > F^M(u^+) \geq Q_M.$$

Thus for large enough N, first is best.

A7.4. Proof of Lemma 1

Working backwards, a consumer's continuation value of searching C given her current best value is v is

$$u_C(v) = -s + x(v+\delta) + (1-2x)v + x(v-r) = -s + v + x(\delta - r),$$

reflecting that the consumer returns if $u_C = v - \delta$ because $\delta \geq r$. Thus, when $u_B = v$ then if $\delta \geq r + \frac{s}{x} \equiv \delta_2$ it is best to search and otherwise it is best to accept at B. Else, it is clearly optimal to accept if $u_B = v + \delta$ and to search if $u_B = v - \delta$, the latter because we assumed $\delta \geq r$.

A consumer's continuation value of searching B given her current offer is $u_A = v$ equals

$$u_B(v) = -s + x(v+\delta) + (1-2x)\max(v, u_C(v)) + xu_C(v),$$

reflecting that the consumer accepts $u_B = v + \delta$, either accepts or searches at $u_B = v$, and searches when $u_B = v - \delta$. Plugging in for u_C from above and rearranging terms obtains

$$u_B(v) = \begin{cases} v + xr + (2-x)(x(\delta - r) - s) & \text{if} \quad s \leq x(\delta - r) \\ v + xr + (1+x)(x(\delta - r) - s) & \text{if} \quad s \geq x(\delta - r) \end{cases},$$

which in turn implies that $u_B(v) \geq v$ (and thus that it is optimal so search if $u_A = v$) whenever $\delta \geq \frac{x}{1+x}r + \frac{s}{x} \equiv \delta_1 \leq \delta_2$, else it is optimal to accept. Again, that it is best to accept if $u_A = v + \delta$ and to search if $u_A = v - \delta$ holds by the same logic as for u_B.

This completes the proof other than the implicit assumption in the calculation of $u_B(v)$ above, that a consumer never returns to A immediately after sampling B. However this is assumption is not restrictive given the finding in Janssen and Parakhonyak (2014) that it is never optimal to return until all sellers are visited.

A7.5. Proof of Proposition 6

The proposition follows by simply computing the quantities sold in each of the three regions. In region $\delta \in [\delta_2, \infty)$ in which consumers search until either obtaining the highest payoff $v + \delta$ or reaching the end, the sale probabilities are

$$(Q_A, Q_B, Q_C) = \left(\frac{1}{2}(3 - -2x)x,\ \frac{1}{2}(3 - -4x)x,\ 3(x-1)x + 1\right),$$

in which the consumer mixes with equal chance between A and B in cases in which she is indifferent about returning to either. Inspection reveals that for $x \in [0,\ x_1]$ last (C) is best, and for $x \in [x_1,\ 1/2]$ first (A) is best.

If $\delta \in [\delta_1,\ \delta_2)$ then consumers use threshold $v + \delta$ at A and lower their threshold to v at B. Consequently quantities are

$$(Q_A, Q_B, Q_C) = \big(x(-2x^2 + x + 1),\ (x-1)^2,\ x(2(x-1)x + 1)\big),$$

and again inspection reveals that for $x \in [0,\ x_2]$ second (B) is best and for $x \in [x_2,\ 1/2]$ first (A) is best.

Finally, if $\delta \in [0,\ \delta_1]$ then the consumer uses threshold v at both sellers A and B, leading to quantities

$$(Q_A, Q_B, Q_C) = (1 - x,\ x(1-x),\ x^2),$$

so that first (A) is best for all $x \in [0,\ 1/2]$.

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